

LPT oscilloscope

1. Introduction

For this constructions I was inspired [LPTScope](#) from Bojan Banko. I build a new hardware and modify software with permission from author.

This construction is free for personal use, for commercial use you need permission from author (from author original software too).



2. Technical statements

Supply voltage	~9 - 15V
Input voltage DC/AC (peak to peak)	0,2V; 0,5V; 1V; 2,5V; 5V; 25V
Max. input voltage	100V
Input resistance	1M Ω
Maximal sample rate	theoretical 600kHz practical 300kHz
Vertical shifting of zero	-10% to +60% input range
Setting up of trigger	0 to 100% input range
Connection to PC	bidirectional LPT port

3. Circuit

For A/D converter I use TLC0820 with protection resistors R1-R8. Input circuit created voltage divider 1:5 with switch PR1.

Switch PR3 set up zero for input and switch PR4 set up measure of AC/DC signal.

Input amplifier is created IO2A. Switch PR2 set up amplify 1:1, 10:1 and 25:1. Summary amplifier is created IO2B and shifting zero for measure AC signal.

Potentiometer P2 and IO2D set up trigger.

The oscilloscope use several stabilized DC voltages +5V (IO5), +9V (IO3), -9V (IO4).

4. Mechanical design

For construction is used one side [PCB](#) with several connection wires (layout chart of components). For oscilloscope is used box U-KM26, see figure [1](#) - [2](#) - [3](#).

5. Setting up

First you install supply part inclusive of K3. You check on voltage +5V, +9V, -9V. Now you install IO2 and all components except IO1 and check function. If it is all right, you can install IO1 and resistors R1-R8 (SMD).

You must set LPT1 on bidirectional or ECP or EPP in BIOS and then connect oscilloscope to LPT1 port PC.

6. Software **LPTScope**

Program is freeware, included source code.

If you connect hardware for the first time, you start measurement - press button "Run". You should to see horizontal black line. Line have vertical shifting, when you turning potentiometer P1. When line not shifting, you must set regular reading from A/D converter (press button AD on toolbar) by means of set up number waiting cycles. You turning potentiometer P1 and look at voltmeter. You must see stable data and data must react to turning potentiometer. Regular reading you check up on to signal applied to input.

One record - activate measure, after buffer is full, measure be stoped

TRG INT - internal software trigger

TRG EXT - internal hardware trigger from input signal. You can modify hardware to external trigger

Violet line - shifting zero of input signal. When is negative shifting, negative voltage not display (it display as zero).

Use mouse - when you press left button, you can move mouse and you will see voltage (Y axis) and time and frequency (X axis)

Time base - set scale on axis X

Y base - set scale on axis Y

Oscilloscope input correction - for right display input voltage (input range)

Sampling rate - setting up when set number waiting cycles for reading from A/D converter

Sample buffer - for saving data from A/D converter. Use max 10000 for **oscilloscope** and max 1000000 for **One record mode**

You can see max 32000 points on the screen. When sample buffer is bigger and you want to see whole buffer, you must set bigger Time base = $32000 / (\text{Sampling rate} * \text{Sample buffer})$.

[Download LPTScope version 1.3](#)

[Download source code](#)

7. Components list

Electrolytic capacitors

C1,C2	220 uF/25V
C3,C4,C6,C7	47 uF/16V
C5	100 uF/25V
C23	22uF/16V

Ceramic capacitors

C8, C9,C10,C11,C12,	
C13,C14,C15,C16	100 nF
C18,C17	22 pF
C20	12 pF
C21	47 pF

Plastic capacitors

C19	100 nF/250V
C22	100 pF/250V

Diode

D2,D1	1N4005
D3,D4,D7	BAT48
D5,D6	1N4148
ZD1	BZX83V5,6V
ZD2,ZD3	BZX83V10
ZD4	BZX83V5V1

Integrated circuit

IO1	TLC0820
IO2	TL074
IO3	78L09
IO4	79L09
IO5	78L05

Connectors

K1 CAN25VP
K2 BNC-Z 50RW
K3 SCD-016

Inductors

L2,L1 220 uH

Switches

PR1,PR3,PR4 P-B143
PR2 P-GS373

Potentiometers PC 1221N

P2,P1 10k

Resistors SMD 1206

R1,R2,R3,R4,R5,R6,R7,R8 680

Resistors 0,4W

R9 1k
R10,R11 100k
R12 120k
R13 680k
R15,R20,R21,R22,
R23,R24,R25 4k7
R16 470
R17 56
R18 180
R19 15
R26,R30 10k
R27 15k
R28 22k
R29,R31,R33 10k
R32 8k2

Resistors 2W

R14 10k
